No Eligible Populations: A Pre-Registered Feasibility Census for Trajectory-Level Conformal Risk Control on Public Agent Traces

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Abstract

Conformal risk control (CRC) can in principle certify an agent pipeline's trajectory-level failure risk at a pre-set level α, abstaining on trajectories it cannot certify. Whether that guarantee can be exercised *honestly* on the agent trajectory corpora the field actually has is an empirical question that is usually answered by shopping for a corpus that works. We answer it the other way: we pre-registered a feasibility census — frozen data floors (exact Clopper–Pearson split sizes, a pre-specified base-failure band, task-grouped disjoint splits), a frozen outcome-binarisation rule that refuses post-hoc interpretation, and a frozen eligibility decision table — over the nine public software-engineering agent populations that survive a counted-evidence screen (332,499 trajectories in two artefact families). **No population is eligible.** All six Claude cells of the SWE-smith trajectory collection have base failure rates of 0.545–0.610, above the pre-registered 0.50 ceiling; all three CoderForge source populations carry an outcome value (-1.0, on 0.25–0.84% of rows) that appears nowhere in the artefact documentation and that the frozen binarisation may not interpret. The result is a registered H-DATA outcome: the binding constraints on trajectory-level CRC today are data-side and specific — outcome encodings and base-rate concentration — not the guarantee machinery, which we verify controls risk on planted-signal synthetic traces and which previously ran end-to-end (to a negative verdict) on the one corpus ever attempted. We report the full census, the counterfactual rates that make the cost of the discipline visible, and three recommendations for trajectory-corpus publishers.

1 Introduction

An agent that acts over many steps fails or succeeds as a *trajectory*, and a deployment guarantee worth having is a statement about trajectories: *among the trajectories this system accepts, the failure rate is at most* α — with abstention, not silent risk, as the price of the guarantee. Conformal risk control [Angelopoulos et al., 2022] supplies exactly this shape of statement for a scalar score: choose a threshold on held-out calibration data so that the expected loss is controlled at α with a finite-sample, distribution-free correction. Lifted to agent traces — a per-trajectory risk score, a calibration split, an abstention rule — it is the natural certification layer for agentic systems, and learned per-step scorers in the style of process reward models [Lightman et al., 2023, Wang et al., 2024, Xi et al., 2025] are the natural scores to certify.

What is missing is not the machinery but the evidence that the exercise is feasible on real recorded populations under honest rules. Feasibility here is fragile in a specific way: every degree of freedom in corpus choice — which collection, which subset, what counts as a pass, which base rates are "in range" — can be, and routinely is, resolved after seeing the data, at which point the guarantee's honesty is nominal. The remedy is the one this paper applies: freeze the feasibility criteria *before* touching outcomes, as a pre-registration, and let the census fall where it falls.

Our contribution is that census, executed end-to-end under a frozen registration (itself the third design iteration of a programme whose earlier pilot and design stages are part of the record). Over the nine public populations that survived a prior counting screen on documented evidence, the census finds **zero eligible populations**, for two disjoint, fully documented reasons: base-rate concentration above the registered ceiling in one family, and an uninterpretable outcome encoding in the other. Sections 2–4 give the machinery and the frozen protocol; Section 5 gives the census; Section 6 discusses what the negative result does and does not establish; Section 7 states recommendations for corpus publishers. The paper's claim is deliberately narrow: it is a feasibility result about the present public data landscape for trajectory-level CRC, produced under rules strict enough that a positive answer would have meant something.

2 Background and related work

Conformal risk control. CRC [Angelopoulos et al., 2022] generalises conformal prediction from coverage to expected-loss control: for a monotone loss $L(i, \lambda) = \mathbf{1}\{\text{failed}(i)\} \cdot \mathbf{1}\{S(i) \leq \lambda\}$ in a threshold λ on a score S, choosing the largest λ whose corrected empirical risk $(\sum_i L(i, \lambda)+1)/(n+1)$ stays below α on a calibration sample controls the expected loss at α under exchangeability. We use this threshold rule unchanged; the trajectory-level content is in the score and in the split discipline (Section 3).

Learned step scorers. Process reward models score individual steps of a reasoning trace: trained on human step labels [Lightman et al., 2023] or on automatic process supervision [Wang et al., 2024], and lifted to multi-step agent trajectories by AgentPRM [Xi et al., 2025]. Trace-risk modelling without guarantees exists as well (e.g. SafetyDrift's absorbing-Markov-chain risk model over agent traces [Dhodapkar and Pishori, 2026]). None of these places a distribution-free finite-sample guarantee over the learned score, prices abstention against a pre-registered ceiling, or evaluates the construction as a feasibility question with frozen data floors. Our delta is the guarantee, the abstention pricing, and the feasibility discipline — not a better scorer; had the census found eligible populations, the confirmatory signal would have been a standard automatically-trained step scorer of exactly this family, calibrated by CRC.

Corpora. The census covers the two public collections that survived the programme's counting screen: the SWE-smith trajectory release (trajectories of SWE-agent driven by Claude models on synthetic software-engineering tasks, in three message formats) and CoderForge-Preview (OpenHands trajectories driven by Qwen3-Coder-480B on tasks from three sources), the latter documented by the SWE-Critix construction [Hu and Chong, 2026]. Corpora excluded at the counting stage (HAL slices, Open-SWE-Traces, earlier candidates) are excluded for counted, documented reasons in the programme record and are not re-litigated here.

3 Method

3.1 Trajectory-level CRC (T-CRC)

A trajectory i has steps $1 \dots T_i$ and a binary outcome (failed / passed) under its benchmark's own pass condition. A per-step scorer $s(i, t) \in [0, 1]$ estimates step-level failure propensity; the trajectory score is the hazard aggregation $S_{\text{haz}}(i) = 1 - \prod_t (1 - s(i, t))$, with $S_{\max}$ and S_{mean} as

descriptive variants. The confirmatory pipeline: fit the scorer on a *fit* split; pass a mechanical signal-quality gate on a disjoint *gate* split (the score must beat the trajectory-length baseline $S_{\text{len}}(i) = T_i$ decisively: AUC ≥ 0.65, ΔAUC $\geq +0.10$, bootstrap 95% CI for the difference excluding 0 — a learned signal that cannot beat "how long did it run" has no business being certified); select $\hat{\lambda}$ by the CRC rule on a disjoint *calibration* split; evaluate once on a disjoint *test* split, reporting risk and abstention only as a pair. Because the gate event is measurable with respect to the fit and gate splits, which are disjoint from calibration and test, conditioning on it does not alter the calibration distribution: the guarantee for the gated pipeline is the unconditional CRC statement. A guarantee-alert rule (a test-risk confidence interval lying entirely above α) and an abstention ceiling (40% at $\alpha = 0.10$) complete the verdict logic, frozen as an exhaustive decision table over per-family outcomes.

3.2 The feasibility protocol (frozen)

Eligibility of a population is decided from a *census* alone — artefact-level quantities computed before any signal, gate, or split outcome exists — under these frozen criteria: (i) at least 800 usable trajectories (a step log and a final evaluation both present); (ii) base failure rate in $[0.10, 0.40]$ for the confirmatory tier, $(0.40, 0.50]$ for a descriptive-only tier, outside $[0.10, 0.50]$ ineligible — the band reflects the method's operating point: below 0.10 the class-count rule below cannot be met at feasible sizes, above 0.50 no threshold at $\alpha = 0.10$ can be non-vacuous against a 40% abstention ceiling; (iii) a four-way split realisation (fit 40 / gate 20 / calibration 20 / test 20, by task group, stratified by group outcome composition) meeting the exact Clopper–Pearson floors $n_{\text{cal}} \geq 160$ and $n_{\text{test}} \geq 160$ — $n = 150$ gives a worst-case 95% half-width of 0.0513, $n = 160$ gives 0.0495, and the floor is set so the interval can resolve the $\alpha = 0.10$ decision — and at least 40 successes and 40 failures in each of gate, calibration, and test.

Outcome binarisation as a frozen function. The subtlest criterion is the outcome itself. A population's successes are defined by its benchmark's own pass condition, applied by a rule fixed before the census sees any value: for a boolean field, success is its true value; for a numeric field with observed distinct-value set V, success is "value = 1" if every value in V is exactly 0 or 1, "value exactly 1.0" (a full pass; fractional values are partial passes) if every value lies in $[0, 1]$, and *ineligible* if any value lies outside $[0, 1]$ or is non-numeric — no clause may be selected or invented after V is observed. This is the criterion that decides the CoderForge family, and we flag its strictness as a deliberate design choice: a sentinel value that the benchmark does not document cannot be assigned a meaning by the analyst without converting the census into the post-hoc exercise it exists to prevent.

Splits and leakage. All splits are by task group (every trajectory of a task in exactly one split), because the text features of any learned scorer make task-identity leakage a concrete failure mode when the same task appears on both sides of a split. Task identity itself is a census item (Section 4).

3.3 Programme context

The registration executed here is the programme's fourth artefact: a v1 pilot on a HAL corpus (Section 5.4), a V2 design retired at design review, a V3 design that found no countable corpus at its floors, and this V4 registration over the populations that a dedicated counting pass (artefact-level counts and documented facts only, no outcome values) found standing. The registration was

scored before execution by a fresh design-stage tester (design quality 7/10; venue screen 9/12, low flag, at this journal) and its execution was audited after the fact by a second fresh tester (Section 8).

4 Census protocol

Each population was censused by streaming its artefact files at the revision counted in the prior screen (CoderForge-Preview at revision `060fca96`; SWE-smith trajectories at `08e109b4`), one file at a time with per-file checkpoints, retaining only aggregates. Per population the census records: rows in file vs. usable rows; parse failures (zero in all nine populations); the outcome field's distinct values, verbatim, with counts; the task-identifier field as found and the number of distinct tasks; median and mean trajectory length in steps; and step coverage. SWE-smith populations are (message format × model) cells of one release, filtered on its model field; its format cells share one task pool — the xml and ticks cells for the same model are the same trajectories in different formats, with identical outcome counts — and its cells are therefore never treated as independent corpora.

Task identity. SWE-smith carries `instance_id` as a column. CoderForge carries no task-id column; schema identification at census found the task identity in the artefact's `image` field (the per-task environment image; mechanically verified on one inspected file of 148 rows: `trajectory_id` minus its run suffix equals the image string on 148/148 rows). The final distinct counts — CF-1 4,492 images over 77,169 trajectories, CF-2 444 over 148,001, CF-3 4,274 over 32,964 — show these are per-repository environment images of varying granularity, not per-instance identifiers (near-instance only for CF-3). Grouping by image would have been task-conservative (no task can straddle splits when grouped by a coarser key) but coarser than task grouping. An independent adjudication of this identification, its timing (recorded before any eligibility decision), and its consequences is part of the programme record and is summarised in Section 8: it changes no outcome, since CoderForge's eligibility fails on the outcome encoding under any reading of the task-identity question.

5 Results

5.1 The census

Table 1 is the result. Of 332,499 censused trajectories (the sum over the nine registered populations; the two families partially share the SWE-smith task universe, and SS-2/SS-3 duplicate one another's trajectories across formats), **no population is eligible** under the frozen criteria, and no population reached the split realisation: the sample draw, the split-formation checks, and eligibility leg (iii) were never evaluated for any population, because every population failed at the outcome stage — leg (ii) for SWE-smith, the binarisation rule for CoderForge. The programme verdict under the registration's own decision table (both families without an eligible population) is the pre-registered **H-DATA** outcome: the confirmatory programme does not run, and the census is the finding.

5.2 Barrier 1: base-rate concentration (SWE-smith)

Every Claude cell of the SWE-smith release fails at a base failure rate between 0.545 and 0.610. This is not a threshold artefact of the census: the populations are large (5,751 to 18,457 usable trajectories), the outcome field is an unambiguous boolean (`resolved`), and the rates are what they are — SWE-agent driven by these models resolves roughly two in five of these synthetic tasks. A

Population	Usable	Tasks	Binding quantity	Outcome
CF-1 (SWE-Rebench source)	77,169	4,492	-1.0 on 255 rows (0.33%)	ineligible
CF-2 (SWE-Smith source)	148,001	444	-1.0 on 367 rows (0.25%)	ineligible
CF-3 (R2E-Gym source)	32,964	4,274	-1.0 on 278 rows (0.84%)	ineligible
SS-1 (tool, Claude 3.7)	17,715	9,143	base failure 0.6102	ineligible
SS-2 (xml, Claude 3.7)	18,457	11,824	base failure 0.5447	ineligible
SS-3 (ticks, Claude 3.7)	18,457	11,824	base failure 0.5447	ineligible
SS-4 (tool, Claude 3.5)	5,751	1,965	base failure 0.5921	ineligible
SS-5 (xml, Claude 3.5)	7,091	2,223	base failure 0.5875	ineligible
SS-6 (ticks, Claude 3.5)	6,894	2,222	base failure 0.5863	ineligible

Table 1: The feasibility census. "Binding quantity" is the first frozen criterion each population fails: for CoderForge, the outcome value set contains -1.0, which the frozen binarisation function's third clause makes census-ineligible; for SWE-smith, the base failure rate exceeds the 0.50 ceiling of eligibility leg (ii) (leg (i), usable ≥ 800, passes for all six). Usable counts reproduce the prior counting screen exactly; parse failures are zero in all nine populations. Median trajectory lengths: CF 48–59 steps, SS 15–30 steps.

confirmatory method whose registered operating point is $\alpha = 0.10$ with a 40% abstention ceiling cannot certify such populations without abstaining on nearly everything, which the anti-vacuity discipline of the protocol exists to refuse. The band's ceiling is doing intended work here.

5.3 Barrier 2: an uninterpretable outcome value (CoderForge)

CoderForge's `reward` is a float taking values 1.0 (89,501 + 44,739 + 20,904 rows across the three sources), 0.0, and -1.0 — 255, 367, and 278 rows in CF-1, CF-2, CF-3 respectively. No definition of the -1.0 value appears in the artefact documentation or in the counting and census records we inspected; plausible readings exist (an execution or evaluation failure sentinel is the obvious one), but the frozen binarisation has no clause for a value outside $[0, 1]$, by design, and inventing one at census time is exactly the move the pre-registration prohibits. All three populations are therefore ineligible on the outcome encoding.

Exploratory counterfactuals (unregistered). To make the cost of that discipline visible — and only for that purpose — we record what two unregistered readings of -1.0 would have given. Excluding the -1.0 rows, the CF failure shares are CF-1 0.4183, CF-2 0.3938, CF-3 0.3605. Counting -1.0 as failure gives CF-1 0.4202, CF-2 0.3953, CF-3 0.3659. The band assignment is the same under both readings: CF-2 and CF-3 would have sat inside the confirmatory band, CF-1 in the descriptive band (it did not attain descriptive-tier status, which is a registered status, and we do not describe it as a descriptive-tier population). For CF-2 and CF-3 the binding barrier is thus the outcome encoding, not the band; that is a statement about documentation, obtainable only because the registered rule refused to make it silently.

5.4 Machinery evidence

The bottleneck is not the guarantee machinery. The programme's validation harness (48 checks, all passing) includes the classical anchors — single-turn CRC reproducing its nominal risk (mean test risk 0.1000 at $\alpha = 0.10$), planted -dependence synthetics controlled by the hazard score (mean risk 0.0154) — and a synthetic prefix anchor for the full learned pipeline eventually intended for

the confirmatory runs: on synthetic traces with a planted per-step error signal, the frozen feature map, stacked heads, and hazard aggregation recover the signal (gate AUC 0.995) and CRC controls the risk at $\alpha = 0.10$ (test risk 0.080), at an abstention of 0.420 — above the 40% ceiling, so the anchor demonstrates risk control, not a confirmatory pass, and we report it as such. What the census establishes is that this machinery currently has no eligible natural population to run on.

5.5 The prior pilot, in its place

Under the programme's separate v1 registration, one HAL corpus (tau-bench airline) was run end-to-end with a frozen heuristic signal ladder: base failure 0.84, calibration AUC 0.482, threshold at abstain-all for every α, abstention 100% at $\alpha = 0.10$ — a falsifier-direction result for that signal class on that corpus. It is v1 evidence, cited for the programme's history; the present verdict does not rest on it, and its corpus's 0.84 base rate would itself be ineligible under the present band, which post-dates it and has no retroactive effect on its recorded verdict.

6 Discussion

What the result says. Under criteria frozen before outcomes were seen, the two public software-engineering agent trajectory collections that survive a counted screen cannot support confirmatory trajectory-level conformal risk control: one family's populations all sit above the base-rate ceiling, and the other's outcome field cannot be binarised without inventing a meaning for an undocumented value. Both barriers are properties of the data landscape, not of the guarantee: the same protocol, unchanged, would have run on any population that met it, and its machinery anchors pass. Pre-registration is what makes this a result rather than an excuse: the H-DATA outcome was one of the registration's pre-declared verdicts, reached by its own decision table, with floors and bands that were never adjusted after the data arrived.

What it does not say. It does not say trajectory -level CRC fails on agent traces: no eligible population was ever run, and the one historical run (Section 5.4) tested a heuristic ladder, not the learned signal. It does not say the band is "right" in any absolute sense: the band encodes an operating point ($\alpha = 0.10$, 40% abstention ceiling), and a programme targeting a different operating point would draw it elsewhere — but would have to draw it in advance, as this one did. And it does not generalise beyond its census: two artefact families, one task domain, nine populations.

The cost of the discipline, made visible. The exploratory counterfactuals of Section 5.2 are the honest price tag of the frozen binarisation: two populations that a post-hoc analyst would very likely have run (failure shares 0.36–0.39 under either plausible reading of -1.0) were turned away over 0.25–0.84% of rows whose meaning the artefact does not document. We judge that price correctly paid — a guarantee certified on an invented outcome definition is not a guarantee — but it is a price, and corpus publishers can reduce it to zero for everyone (Section 7).

7 Recommendations

We label these as recommendations; each is tied to the census fact that motivates it, and none is an experimental finding.

1. **Document sentinel outcome values, or encode execution failure separately from task failure.** An undefined -1.0 reward value, present in all three CoderForge sources at

0.25–0.84% of rows, is the sole reason three countable populations could not be binarised under a pre-registered rule.

2. **Ship the benchmark's pass/fail outcome as a boolean field.** SWE-smith's `resolved` boolean was the one outcome encoding in this census that required no interpretation whatsoever.

3. **Carry task/instance identifiers as a column.** SWE-smith's `instance_id` column made task-grouped splitting unambiguous; CoderForge has no task-id column, and its environment-image field is a per-repository identifier whose granularity varies by source (444 to 4,492 distinct values over 33,000–148,000 trajectories).

8 Limitations

The census covers two artefact families in one task domain (software-engineering agents); other domains' corpora were excluded at the programme's counting stage for documented reasons and are outside this paper's claims. The CoderForge task-identity identification (Section 4) rests on mechanical verification of one inspected file and an adjudicated interpretation; it is outcome-immaterial here but would matter in any future run on this artefact, where per-instance identifiers would have to come from the publisher or from the trajectories' own content under a new registration. The exploratory rates of Section 5.2 are unregistered counterfactuals, reported for interpretation; neither reading of -1.0 is endorsed. The synthetic machinery evidence (Section 5.3) uses planted signals and demonstrates control, not confirmatory performance. Finally, the eligibility band and floors are design choices: we argue for them from the method's operating point, and other pre-registered choices are possible — what is not optional, on this paper's argument, is making them in advance.

Falsifier bookkeeping. The programme sits inside a larger area whose pre-registered falsifier requires non-vacuous trajectory-level control (at most 40% abstention at target risk) on at least two of three trace corpora: a HAL task family, own-lab traces, and METR runs. The HAL family carries the v1 falsifier-direction result; the other two are untested; the falsifier has therefore neither fired nor been passed. The present census's families are none of those three, and its results count neither for nor against that falsifier. H-DATA is this programme's verdict under its own registration; any area-level kill decision is a separate, recorded act by the programme's owner, not a consequence this paper asserts.

Reproducibility. Every number in Table 1 recomputes from the census records produced by the programme's census code over the two artefacts at the revisions named in Section 4. On freeze integrity, precisely: the programme's earlier registration chains (V2, V3) and the present registration chain's non-registration files verify against their recorded hashes today; the present registration's Part I was hash-verified at its pre-execution design-stage check, and its Part II execution record was appended under the registration's own rule. The code, registrations, census records, and tester reports are deposited in a public repository: `https://github.com/ranadhir19/trajectory-conformal-census`.

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